

AniME: Adaptive Multi-Agent Planning for Long Animation Generation

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Abstract

We present **AniME**, a director-oriented multi-agent system for automated long-form anime production, covering the full workflow from a story to the final video. The director agent keeps a global memory for the whole workflow, and coordinates several downstream specialized agents. By integrating customized Model Context Protocol (MCP) with downstream model instruction, the specialized agent adaptively selects control conditions for diverse sub-tasks. AniME produces cinematic animation with consistent characters and synchronized audio-visual elements, offering a scalable solution for AI-driven anime creation.

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1 Introduction

Anime production is a complex, labor-intensive process evolves multiple stages such as scriptwriting, storyboarding, character and scene design, animation, voice acting, and final editing. Traditional

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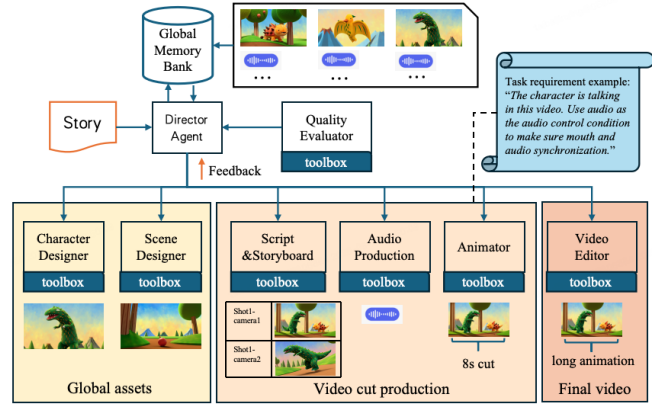


Figure 1: Brief Architecture of AniME

workflows require extensive manual expertise and close collaboration across diverse teams, resulting in high costs and long production cycles. Recent advances in generative AI, such as AniSora [Jiang et al. 2024] for animation generation, have demonstrated impressive capabilities in specific tasks [Du et al. 2025; Majumder et al. 2024]. However, these methods each exhibit distinct strengths and weaknesses in particular domains, leading to challenges in maintaining consistency [Shi et al. 2025; Yang et al. 2024] and achieving fine-grained controllability in agent-driven video generation [Li et al. 2024; Wu et al. 2025; Xia et al. 2025]. As a result, developing a fully automated system for long-form anime generation remains an open challenge, particularly in selecting appropriate control conditions and ensuring cross-stage content consistency.

In this work, we present *AniME*, a novel director-oriented multi-agent framework that integrates specialized agents equipped with

Table 1: Summary of Specialized Agents and their MCP Toolsets

Agent	Inputs	Outputs	Tools & Methods
Character Designer	visual style, character description	character multi-view image	Text-to-image + refinement; multi-view synthesis
Scene Designer	visual style, Scene description	Background images, layered assets	Depth-guided image generator; layout-guided image generator; relighting model
Script & Storyboard	Shot description text	Timeline, shot prompt, keyframes, camera motion	LLM-based segmentation and tagging; camera planner; layout planner; reference image generation
Animator	Keyframes/camera paths/rigs/poses/audio	Frame sequences	Keyframe/audio/pose/camera conditioned video generation model
Audio Production	Dialogue with emotion labels, scene tags	Phoneme-aligned audio	speaker conditioned TTS; text/video to music model; audio mixer programs
Video Editor	Frame sequences, audio stems, editorial instructions	Final encoded video	Transition effects; color pipeline; FFmpeg multi-pass encoding
Quality Evaluator	Generated frames and assets	text-to-video similarity	text-to-image similarity; identity verification; AV sync checks; VLM narrative evaluator

tailored Model Context Protocol (MCP) [Anthropic 2024] toolsets. The framework employs centralized planning and quality control to coordinate task scheduling and ensure content consistency across stages. It enables coherent long-form video generation through a combination of rich, controllable generative models and iterative feedback workflows. Furthermore, we customize the MCP protocol of downstream generative tools to explicitly annotate each tool’s domain expertise and limitations, allowing agents to select appropriate control condition models for specific sub-tasks.

2 Method

AniME decomposes the story-to-video task into hierarchical stages coordinated by a centralized **Director Agent** and executed by a set of **Specialized Agents**. Each agent A_i has a well-defined input type I_i , output type O_i , and a local MCP toolbox $\mathcal{T}_i$.

2.1 Director Agent

The Director Agent serves as the central controller of the AniME framework, managing both the global workflow and quality assurance. It decomposes the input story into a task workflow, allocates subtasks to agents, checks the quality of each specialized agent, and maintains a global Asset Memory Bank.

ALGORITHM 1: AniME: Multi-Agent Workflow

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Input: Story script  $\mathcal{R}$ 
Output: Final video  $\mathcal{V}$ 
1 Initialize Director with  $\mathcal{R}$ , derive shots  $\mathcal{H}$  and styles  $\mathbf{s}_v, \mathbf{s}_a$ ;
2 Generate task list  $T$  and task dependencies graph  $\mathcal{W} = (N, E)$ ;
3 foreach task  $t \in T$  in topological order of  $\mathcal{W}$  do
4   Assign  $t$  to agent  $A_i$ ;
5   Agent  $A_i$  select proper model from MCP toolset  $\mathcal{T}_i$  according
    to task requirement;
6   Agent  $A_i$  producing output  $O_i(t)$  and return to Director;
7   if Evaluated quality of  $O_i(t)$  low then
8     Request revision
9   end
10  Store  $O_i(t)$  in Asset Memory;
11 end
12 Director performs final editing via Video Editor Agent;
13 return  $\mathcal{V}$ ;

```

Table 2: Names and key fields for the Asset Memory Table

Table Name	Keys
shot	id, description
scene	id, prompt, view_3d
character	id, prompt, demo_voice, voice_prompt, 3d_view
style	id, visual_style, acoustic_style
storyboard	id, prompt, image_path
video	id, prompt, video_path, shot_id, music_id
music	id, character_id, prompt, music_path

2.1.1 Workflow of the Director. Given a long-form story $\mathcal{R}$, the Director initiates a hierarchical breakdown into scenes and shots through a segmentation process, and decides the style for visual $\mathbf{s}_v$ and $\mathbf{s}_a$ acoustic setting. Then it uses chain-of-thought prompts to generate an initial task list T .

The Director maintains a workflow graph $\mathcal{W} = (N, E)$, where each node $n \in N$ corresponds to a production task (e.g., “generate character pose for shot 3”) and edges E encode explicit dependencies (e.g., *character design* $\rightarrow$ *storyboard* $\rightarrow$ *animation*).

2.1.2 Asset Memory Management. The Asset Memory Management module stores and organizes key creative assets across the production pipeline, ensuring consistency and reusability. The assets are shown in Table 2. The management is implemented using queryable database tables, enabling efficient indexing, retrieval, and update operations.

2.2 Specialized Agents and MCP Tools Selection

AniME employs a set of specialized agents designed for a key production stage with dedicated MCP toolsets. The Table 1 demonstrates a detailed overview of each agent’s inputs, outputs, and tools. Specifically, we describe the advantages and disadvantages of each tool in the MCP protocol, enabling the agent to decide on the task requirements.

The *Script & Storyboard* transforms narrative scripts into structured shot descriptors and visual keyframes with camera plans. The *Character Designer* generates canonical character assets, including multi-view portraits and identity embeddings. The *Scene Designer* creates backgrounds and environmental assets consistent

with scene perspective and style. The *Animator* synthesizes animated frame sequences and performs precise lip-syncing, maintaining character identity and style coherence. The *Audio Production* handles expressive speech synthesis, ambient sound generation, music composition, and audio mixing. The *Video Editor* integrates visual and audio components, applies color grading and transitions, and produces the final encoded video. Finally, the *Quality Evaluator* automates multi-modal verification of visual and narrative coherence, triggering targeted revisions when necessary.

Meanwhile, animation creators can interact with the agent at any stage, guiding the agent to produce animations that better align with the creator’s artistic style.

3 Conclusion

We presented *AniME*, a director-oriented multi-agent framework for automated long-form anime production. By integrating specialized agents equipped with customized MCP toolsets, the agents

adaptively decide the optimistic available models and achieved consistent generation from a story to the final video.

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